KLAi

What data science is and how to think in it

What does a data scientist do, and what will you be able to do after 24 sessions?

AI and Data Science Professional | Session 1 of 24 | Module A: Think with data

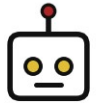
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Name the four roles on a data team, and the six phases of the CRISP-DM lifecycle from business question to deployment

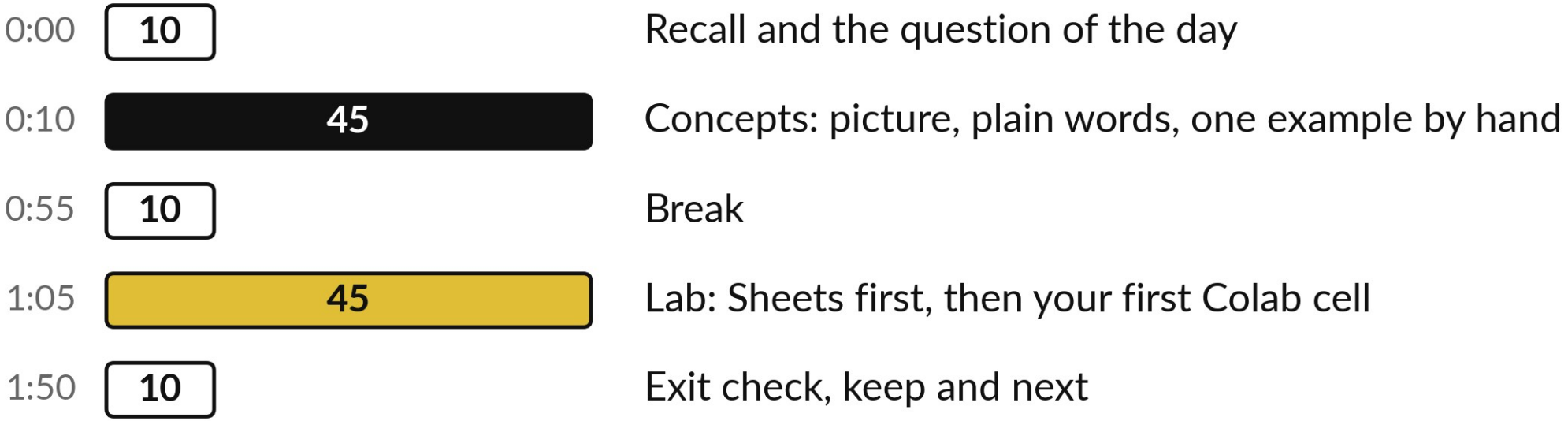


Tell artificial intelligence, machine learning, deep learning and data science apart by what each one produces



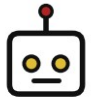
Count the records and variables of a table in Google Sheets, then open Google Colab and run your first cell

Your two hours: concepts first, then a Sheets and Colab lab

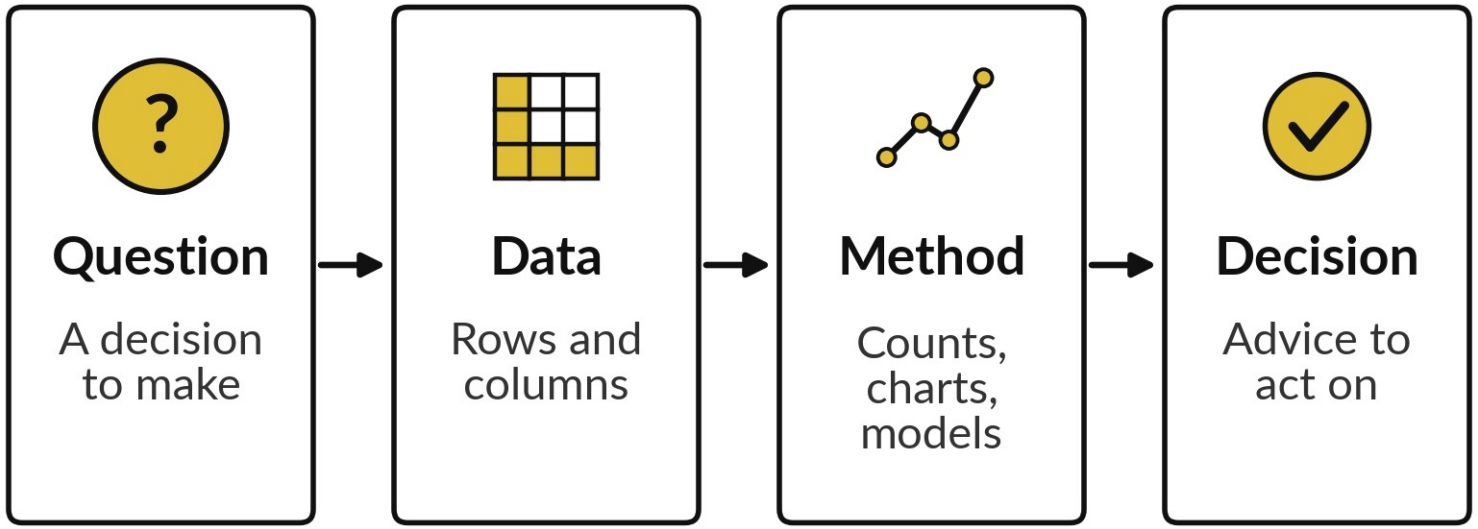


Ideas with pictures first. Then the sheet and a notebook. Then what you keep.

Words we use today

	Word	What it means
	**Data science**	Using data, tools and careful thinking to answer a question and support a decision. It is a way of working that runs from question to decision.
	**Machine learning (ML)**	A computer finds a rule from examples. A person does not type the rule.
	**Dataset**	A collection of data about one subject, arranged in tables. In Excel words: a workbook in which each tab is one table.
	**Model**	A rule found from data that turns inputs into a prediction or an explanation.
	**Lifecycle (CRISP-DM)**	The phases a data project passes through. CRISP-DM is the industry reference: six phases, from business understanding to deployment.

Data science turns a question and data into a decision



Data science is a way of working: someone asks, data answers, and a person decides.

Dataset: a workbook in which each tab is one table.

Model: a rule found from data. You fit your first model in Session 13.

A chart nobody uses has failed.

A business question worth asking names the decision it feeds

A weak question

Tell me about sales.

No decision attached.

No number to find.

A strong question

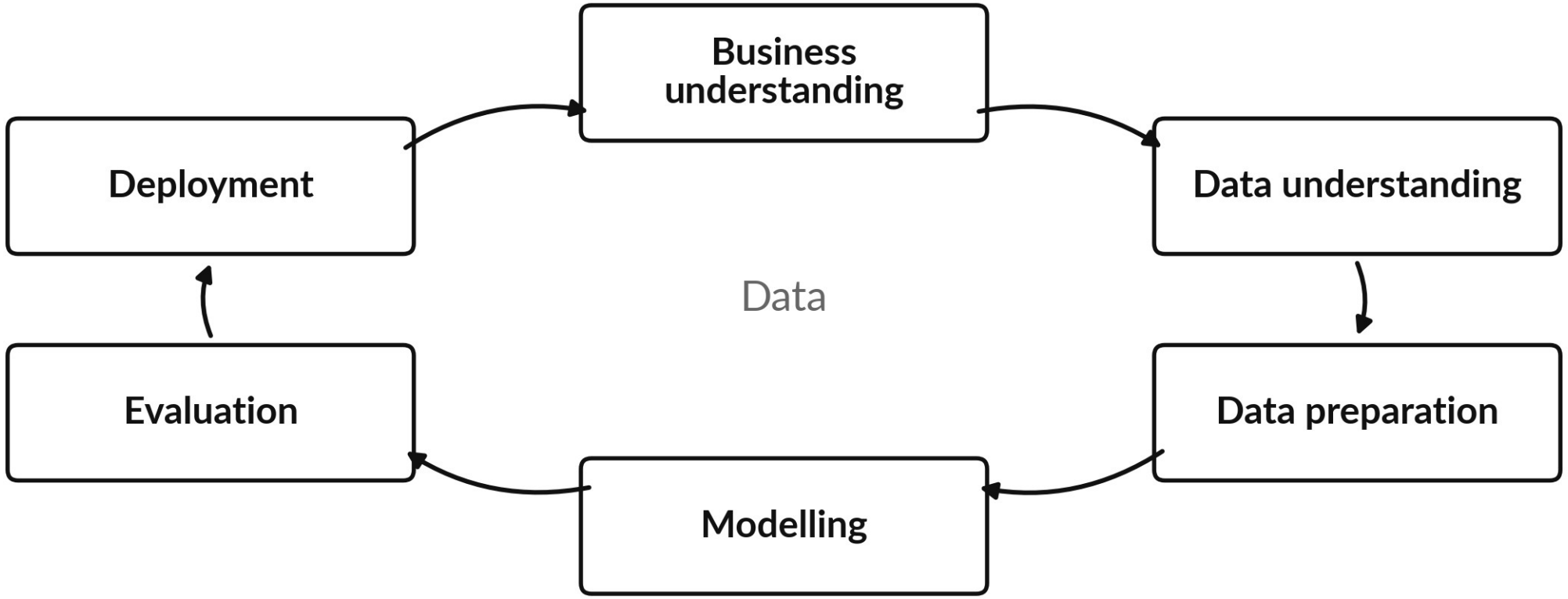
Which branch has the lowest median order value this quarter, so it gets next month's support visit?

It names the measure (median order value) and the decision (the support visit).

The Lakeside workbook has the data.

Write the decision first, then the question.

CRISP-DM, the industry reference, splits a data project into six phases



CRISP-DM: Cross-Industry Standard Process for Data Mining. A later phase can send you back.

Each CRISP-DM phase answers a plain question, and this course covers all six

CRISP-DM phase	The plain question	Where in this course
Business understanding	What decision are we helping with?	Session 1 and the course project, Session 24
Data understanding	What does each table and variable mean?	Sessions 2 to 4 and Session 11
Data preparation	How do we clean and join the data?	Module B, with the cleaning milestone in Session 8
Modelling	Which rule fits the data?	Sessions 13 to 22
Evaluation	Is the result good enough and fair?	Inside every modelling session and Session 19
Deployment	How do other people use it?	Session 23 (a small app) and Session 24

Many teams add a seventh step after deployment: monitor, which means check later that it still works.

One Lakeside question walks through all six phases

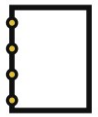
CRISP-DM phase	The Lakeside example
Business understanding	Decide which branch gets next month's promotion budget.
Data understanding	The orders tab has 2,000 records. The branches tab has 8 records.
Data preparation	580 orders have a blank customer_id. They are walk-in shoppers, so keep them.
Modelling	A pivot table of orders and total sales for each branch. A predictive model comes in Module D.
Evaluation	Jinja leads on the mean order, 71,914 UGX. One order of 1,935,520 UGX adds about 7,300 UGX to that mean, but Jinja still leads without it. Check the median and the count too.
Deployment	One page for the manager that shows the count of orders behind each figure.

Thinking in data means four habits, and you practise each one in this course

Before you touch a number, ask these four things.



What decision is this for?



What does one record stand for?

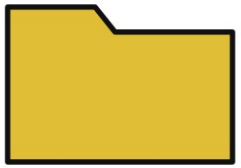


Where did the data come from, and how many records are there?



Did I calculate this number, or did someone tell me?

Four roles share the work on a data team



Data engineer

Builds pipelines and databases.



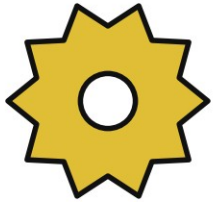
Data analyst

Answers questions about the past.



Data scientist

Tests ideas and builds models.

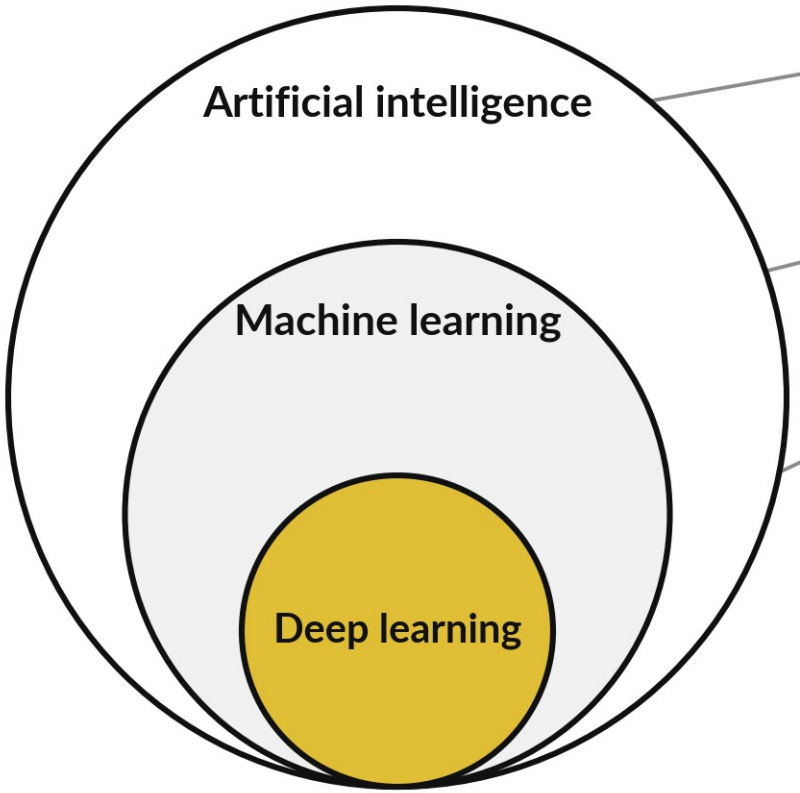


Machine learning engineer

Puts a model into a product.

One person often does several jobs. You practise the analyst and scientist jobs.

Artificial intelligence, machine learning, deep learning and data science differ



Software that does tasks we call smart
Example: a spam filter.

Rules learned from examples
Produces a model. Example: loan default.

Neural networks with many layers
Example: speech recognition.

Data science: a way of working from question to decision. It produces advice, and uses any circle only when it helps.

Data science is the way of working, and often needs no machine learning.

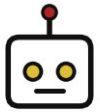
A formula is a rule you write; a model is a rule found from data



Rule written by a person

Revenue = quantity × price × (1 − discount).

It holds on all 6,590 order lines.



Rule found from examples

Nile Microfinance: 8,000 loans, 1,004 defaulted.

An algorithm finds which loan details go with default.

A model is a rule found from data.

Lakeside Mart is the real data for the whole course



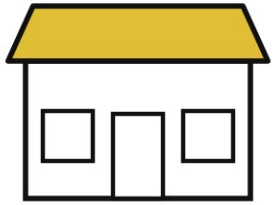
2,000 orders

One record per till order



6,590 order lines

One per product on an order



8 branches

Eight shops in Uganda



150 products

One record per product

Fictional, generated data. A sample of a 30,229-order file.

A chat assistant can state a figure that nobody calculated

Unchecked

You ask for the share of orders paid by Mobile money.

It answers with a clean percentage.

It predicts likely words and may not calculate.

Checked

One COUNTIF formula in Sheets gives it.

976 of 2,000 orders: 48.8 percent.

Write the source beside the figure.

Recompute every number you did not calculate yourself.

Every tool in this course is free for you and for the tutor

	Tool	What you use it for	Cost
	**Google Colab**	Python notebooks in your browser	Free with a Google account
	**Google Sheets (or Excel, LibreOffice)**	Tables, pivot tables and charts	Free, or what you own
	**pandas, NumPy, scikit-learn, matplotlib**	Data work, models and charts in Python	Free, open source
	**SQLite**	Questions to a database with SQL	Free, inside Python
	**GitHub**	Save and share your work	Free account
	**Streamlit Community Cloud**	A small web app for your model	Free for public apps

At the time of writing these are free. Free plans can change, so check current terms before you rely on one.

The course is six modules, and each builds on the one before

	Module	Sessions	What you do
	**A Think with data**	1 to 4	Sheets, no code
	**B Python for data**	5 to 9	Python, pandas, first SQL
	**C Analyse and explain**	10 to 13	SQL, charts, inference, regression
	**D Machine learning**	14 to 19	Build, test and judge models
	**E Neural networks and text**	20 to 22	Networks, text, embeddings
	**F Ship it and present**	23 and 24	A small app, then your course project

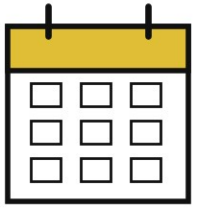
Project milestones are in Sessions 8, 13, 19 and 24. Every session has a concepts half, then a lab where you work.

Colab is free, with limits to plan for



Idle time

A session ends when you stop working.



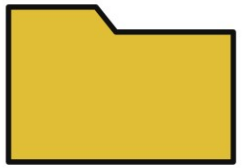
Length

A maximum session time applies.



Quota

Limits are not published and can change.

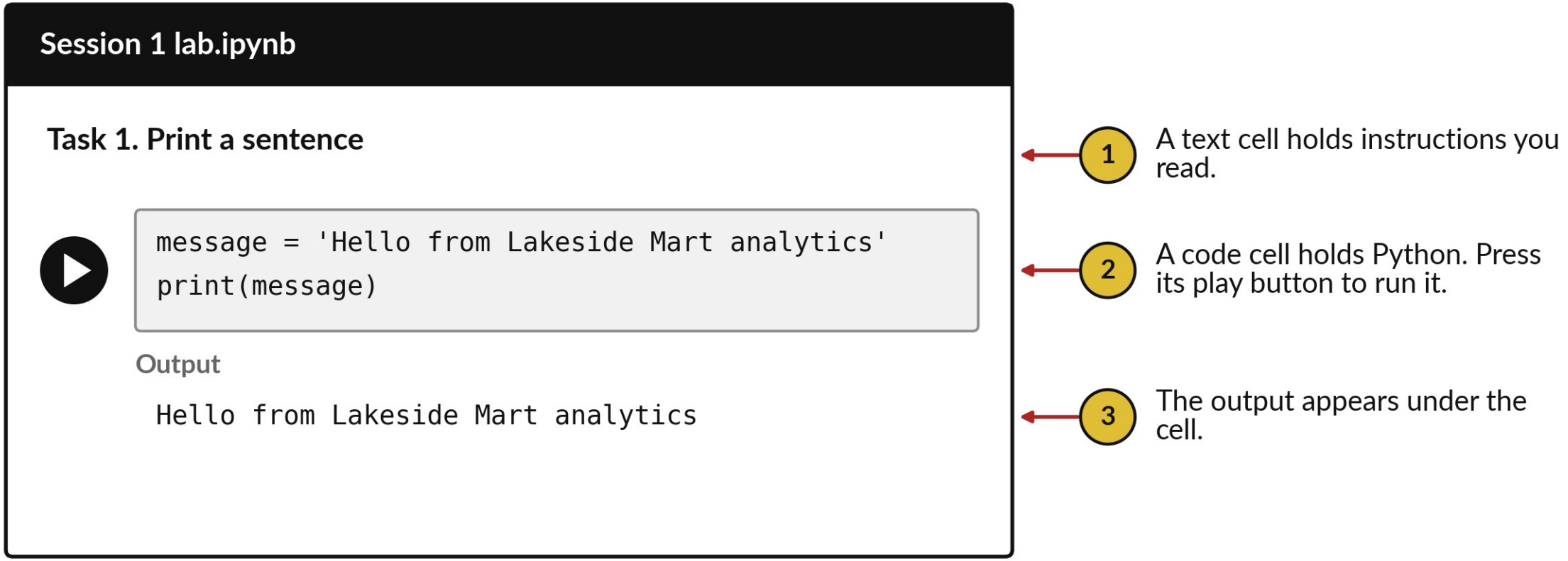


Save

Keep notebooks in Drive or GitHub.

My understanding at the time of writing. Check current limits.

A notebook is made of cells: text cells you read and code cells you run



Python is the programming language. A cell is one box of code. Run it, read it, change it, save it.

Where you will use this at work: frame the question before you open the data

Your job	The question you start from	What today lets you do
Credit officer at a bank in Kampala	Which loan applications are likely to go unpaid?	Write the decision first, then find the loans table.
Monitoring and evaluation officer at an NGO	Did the training raise farmers' yields?	Name the data you need before you promise a report.
Operations manager at a retail chain	Which branch has the lowest median order value, so it gets next month's support visit?	Check any figure an assistant gives you against your own workbook.
Health records officer at a district office	Which months do clinics run short of drugs?	Use the six phases as a checklist for a monthly report.

Read the recipe before the lab: four lines make your first cell

```
message = 'Hello from Lakeside Mart analytics'
print(message)
total = 4500 + 2800
print(total)
```

```
Hello from Lakeside Mart analytics
7300
```

In plain words

Line 1 stores a sentence in a name called message.

Line 2 prints it under the cell.

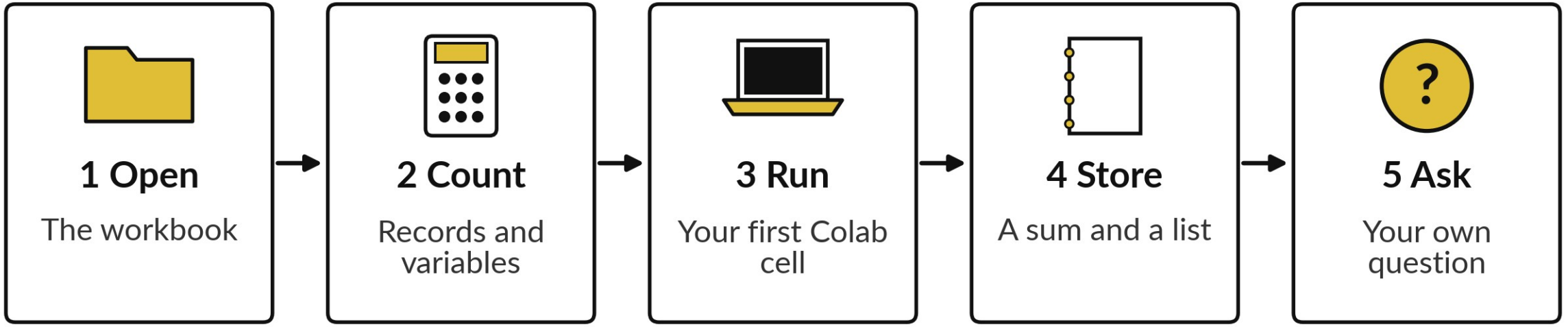
Line 3 adds two illustrative prices in UGX, rice 4,500 and sugar 2,800, and stores the sum in total.

Line 4 prints 7300.

Part 2: the Sheets and Colab lab

Close the slides. Open the sheet.

The lab has five steps: two in Sheets and three in Colab



About 45 minutes. For each task, predict first, then run it, then read the answer. Save your notebook to your Drive at the end.

Check your numbers before you move on

What you compute	You should get
Tabs in the workbook, with READ ME	7
Records in orders, and in order_lines	2,000 and 6,590
Variables in orders, and in order_lines	7 and 7
Blank customer_id in orders	580
Share of orders paid by Mobile money	976 of 2,000, which is 48.8 percent
Output of your first cell	Hello from Lakeside Mart analytics
The sum of 4500 and 2800	7300
Records in the full orders.csv file	30,229

If your number differs, find the task that went wrong. Do not change a number to match.

Five likely mistakes in the lab and how to fix them

What you see	What it means	What to do
A count of 2,001 where you expected 2,000	The range includes the header cell in row 1.	Start the range in row 2: A2:A2001.
NameError: name 'message' is not defined	You ran a cell before the cell that creates the name.	Run the cells from the top, one after the other.
SyntaxError: unterminated string	A quote mark is missing at one end of the text.	Put the same quote mark at both ends: 'text'.
FileNotFoundError: orders.csv	The setup cell has not run, or data.zip was not chosen.	Run the setup cell and choose data.zip.
Runtime disconnected	A free session timed out.	Reconnect, run from the top, and save a copy to Drive.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**In CRISP-DM, which phase comes straight after business understanding, and what do you do in it?**	Data understanding: find the tables that could answer the question and check what they mean.
	**In Excel words, what is a dataset, and how does a model differ from a formula?**	A dataset is a workbook in which each tab is one table. A model is a rule found from data, where a formula is typed by a person.
	**An assistant gives you the share of orders paid by Mobile money. What do you do before you report it?**	Recompute it from your own data: 48.8 percent, 976 of 2,000. Write the source beside it.

Save your notebook and your question, then bring one question to Session 2

What you keep from today

Keep your saved Session 1 notebook in Google Drive, and your one-sentence business question with the decision it would feed.

Our question today

What does a data scientist do, and what will you be able to do after 24 sessions?

Next session

Session 2: Data as tables: records, variables and keys. We open every tab of the Lakeside workbook and learn how the tables connect.

Take away

Your lab file

A saved Colab notebook

Your question

One decision, one question

Next session

Data as tables